P R a V A H

Predictive Routing and Vehicle’s Augmented Handling

Team SATVA

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The Problem: **Inefficient & Reactive Traffic Control in Indian Cities**

India's traffic infrastructure relies heavily on static, fixed-time signal systems that do not adapt to real-time road conditions. As a result, intersections fail to handle sudden congestion, slow emergency vehicles, and cannot mitigate disruptions such as accidents, rallies, floods, or animal crossings. Current systems can monitor traffic but do not respond or optimize traffic flow.

Key Issues

- Signals do not adapt to traffic density, direction, or special cases
- Slow ambulance and emergency response due to poor prioritization
- No automatic action for disruptions (accidents, rallies, animals, waterlogging)
- High time, fuel, and productivity loss due to long idle times

Impact

- 30% of trauma deaths caused by delayed ambulance access (AIIMS, 2021)
- India loses ₹1.47 lakh crore annually due to congestion-related delays & fuel waste (UN ESCAP, 2023)
- 1.1M+ signalized junctions still use outdated fixed-time systems (MoRTH, 2023)

Urban India's Mobility Congestion

India needs an intelligent, adaptive traffic control system that predicts disruptions and dynamically responds to them in real time.

Idea

PRaVAH is a real-time, responsive traffic control system that uses multi-sensor fusion and AI to monitor, predict, and optimize traffic signals automatically.

Key Highlights

- Live surveillance + adaptive signal control
- Optimized even for low-visibility & night time surveillance
- Fully controlled by traffic authorities (RTO / Police / Municipality)
- Integrated alert system for road anomalies

Automatically Detects

- Emergency vehicles
- Accidents & stalled cars
- Hazardous driving & road conditions

PRaVAH model deployment (idea)

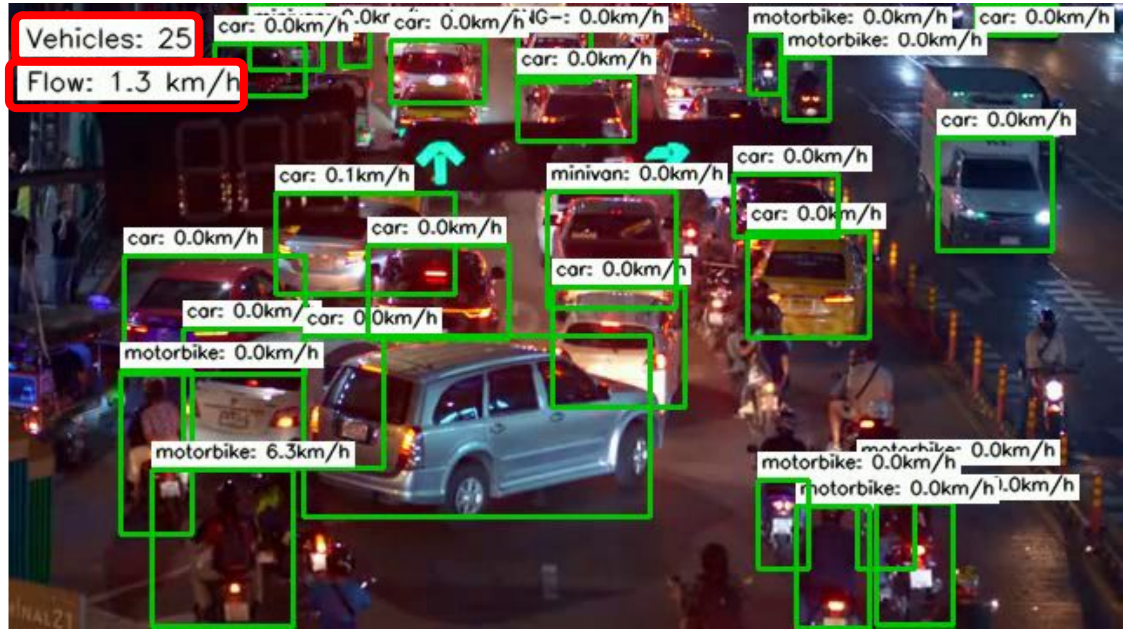
How PRaVAH Works at a Junction:

A compact pole-mounted unit with **stereo cameras + radar** analyzes traffic in real time and **optimizes signals using AI.**

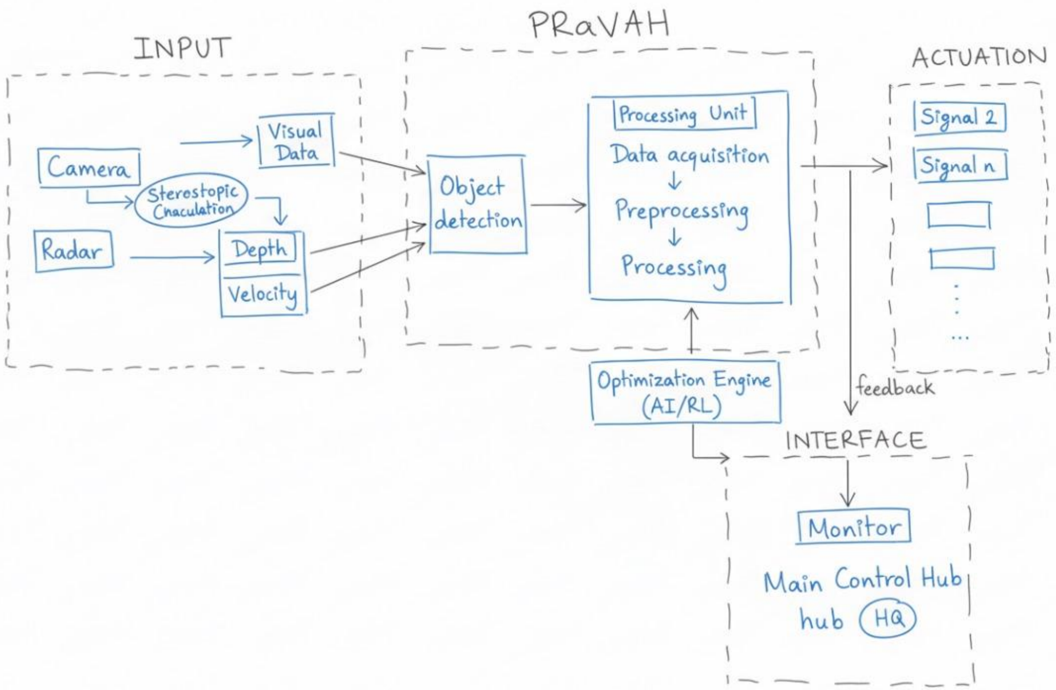
Core Functions

- Vehicle & pedestrian detection
- Speed & density estimation
- Emergency & anomaly detection
- Dynamic traffic flow optimization

Sense → Predict → Alert → Optimize



Data-Driven Smarter Traffic Solution



Workflow

The Size and Nature of Market :

PRaVAH targets authorities responsible for traffic management, emergency mobility, and smart city operations. The demand is driven by rapid urbanization, congestion losses, and the need for AI-based adaptive control instead of static signals.

Major Market Segments

- Urban & Semi-Urban Municipalities - Need to upgrade to adaptive signals to reduce congestion and travel time.
- Traffic Police / Control Centres - Require live dashboards with automated alerts for actionable decision making.
- Emergency Services (Ambulance/Police/Fire) - Need priority routing to reduce delays and save lives.
- Smart City Mission & Infrastructure Bodies - Integrating IoT-based traffic automation at scale.
- Highway/Toll Authorities - Require predictive control to prevent choking and accidents.

Market Opportunity

- India’s traffic signal controller market is valued at USD 76.5 million (2021) and projected to grow to USD 271.8 million by 2029 (CAGR ~17.7%)
- Current adaptive controllers cost ₹40–50 lakh per junction, but PRaVAH delivers similar intelligence at < ₹30,000 per junction → 99% cost reduction, scalable to Tier-2/Tier-3 cities.
- Over 1.1M+ signalized junctions in India still use fixed-time outdated systems, presenting a massive retrofit market.

Customer Validation

We discussed key requirements with *a faculty member from the Civil Engineering Department (Traffic Engineering domain)*

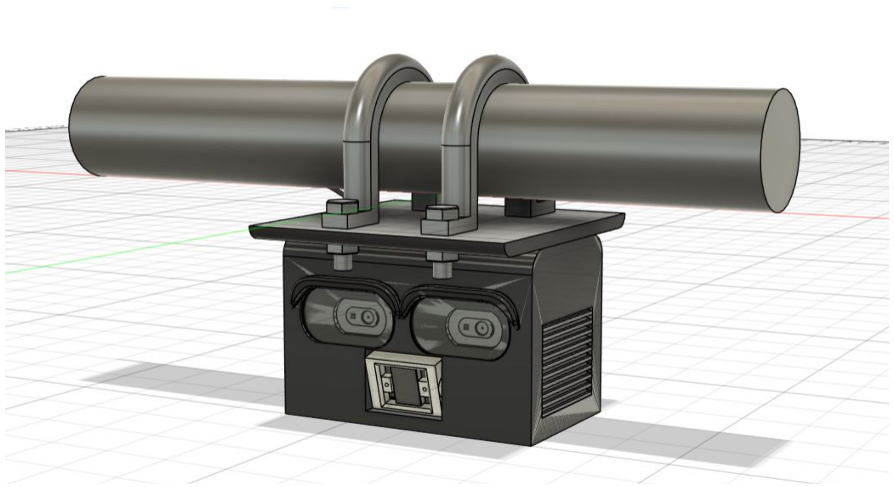
- Automated emergency prioritization and anomaly-aware control are missing in current deployments.
- Low-cost sensor-based retrofitting, rather than full system replacement, is preferred by municipalities.

Work Done So Far

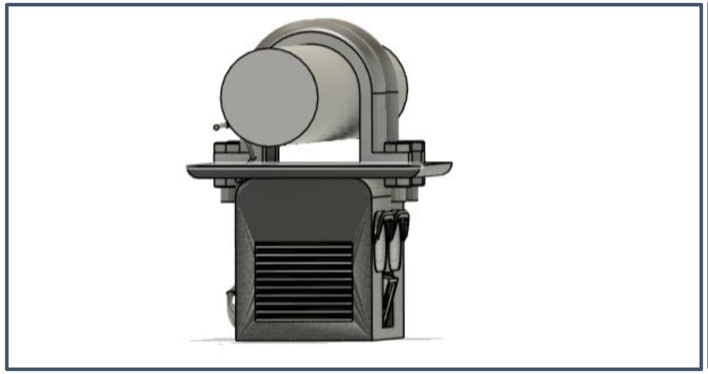
Prototype Development

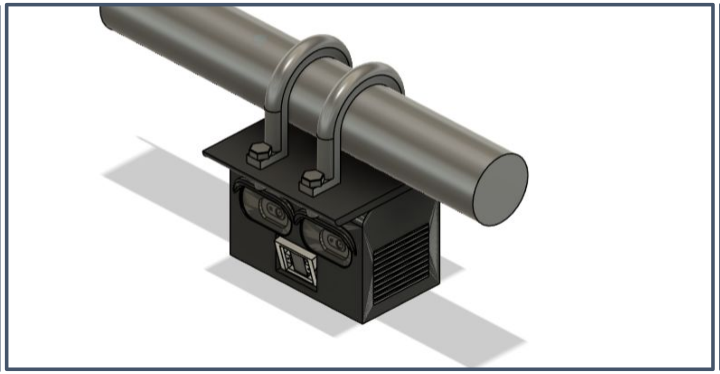
- Designed a pole-mounted enclosure prototype for PRaVAH containing stereo cameras, radar module, and edge processor (CAD model completed).

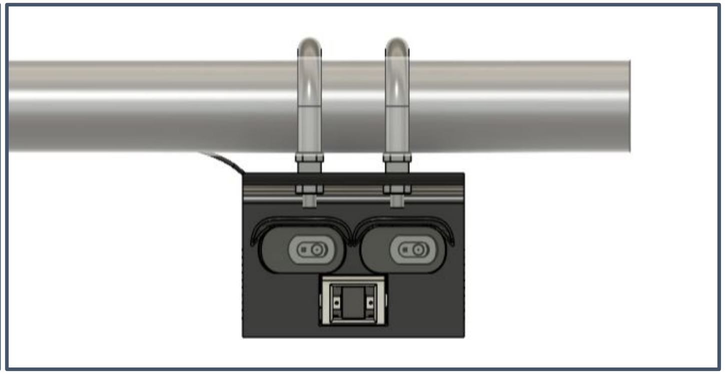
Key Prototype Feature:
The enclosure enables easy mounting on existing poles, requires no change in traffic lights, supports plug-and-operate installation, and allows quick maintenance access.



Prototype Model (CAD)

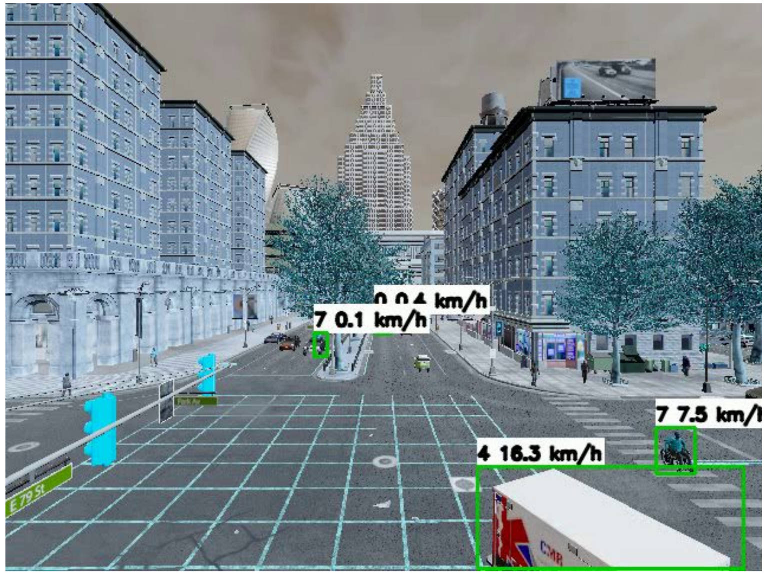






Dataset Simulation and Model Training

- Developed a real-time detection model for vehicles, pedestrians, anomalies, and emergency vehicles using open datasets.
- Built simulation environments to test congestion, emergency routing, and anomaly events safely before deployment.
- Created an authority dashboard to display live alerts, congestion maps, and emergency prioritization controls.
- Integrated backend processing with the dashboard to visualize real-time AI output.
- Collected feedback from a faculty expert in traffic engineering to refine deployment, detection requirements, and signal priorities.



Environment Simulation in Carla

Results of model training

AI-Driven Traffic Monitoring: Initial SITL Simulation Results

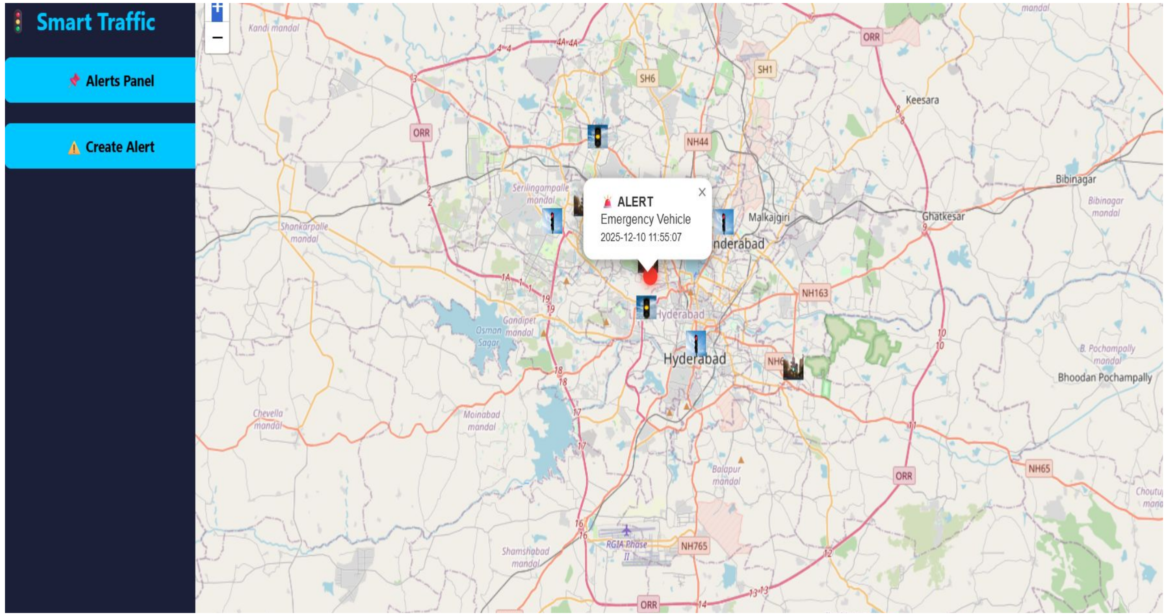
AI Detection & Prediction (SITL Testing)

- Model successfully identifies vehicles, pedestrians, congestion build-up, and traffic flow direction across different densities.
- Detects emergency vehicles and reports anomalies such as accidents, stalled vehicles, unusual crowding, and animal movement.
- Forecast model predicts queue build-up and required green-time adjustments using traffic pattern logs.

Dashboards & Alerts

- System generates alerts for:
  • Emergency vehicle presence
  • Road blockages due to crowding, animals, or accidents
  • Abnormal vehicle motion (stalling or rash movement.
- Provides recommended actions: dynamic signal extension, emergency corridor creation, adaptive pedestrian crossing.

Inference: System can monitor, detect, and recommend signal actions in real time.



DASHBOARD SHOWING ALERT OF AN EMERGENCY VEHICLE

Work planned till Finale:

Hardware Integration and Testing

- Install and calibrate stereoscopic cameras, radar module, and edge processing unit on prototype enclosure.
- Test communication latency between device, signal controller, and dashboard system.
- Train reinforcement learning engine for real-time optimization using recorded traffic data.
- Improve detection reliability under low-visibility conditions such as rain, fog, or night-time.

Deployment Preparations

- Conduct human-in-the-loop deployment test at a controlled junction (e.g., IIT Gate / Simrol).
- Complete authority dashboard with alert system for police and municipal controllers.
- Evaluate performance metrics including:
  - response time
  - emergency clearance delay
  - congestion reduction and green-time efficiency

Goal for Finale: Fully functional device with optimized traffic response demonstrated at a live or controlled junction.

An Idea for Smart India

Challenges

- **Limited access to ground truth data**
  Lack of labeled traffic data from local intersections makes training difficult and simulation-dependent.

- **Model fine-tuning and confidence thresholds**
  Detection accuracy for emergency vehicles, animals, and anomalies requires dataset balancing and threshold calibration to minimize false-positive alerts.

- **Adaptation to unpredictable road conditions**
  Real roads involve non-lane behavior, overcrowding of two-wheelers, roadside vendors, temporary blockages, and unplanned diversions which make it difficult for early-stage models to generalize.

- **Wide system scope vs. problem specificity**
  PRaVAH could support many features (driver behavior, routing, pollution, audit trails), but to fit the problem statement we had to narrow the solution without narrowing its long-term utility.

Challenges

- **Night-time and adverse weather**
  Fog, low illumination, glare, shadows, and rain can degrade camera performance.

- **Hardware deployment permissions**
  Final installation requires involvement of Traffic Police, RTO, Smart City officials, and other relevant departments. But their collaboration timelines are highly uncertain.

- **Cost-effective hardware optimization**
  Radar + stereo vision + edge compute must remain highly cost optimised for each junction without losing accuracy or night-time reliability.
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- **Feedback and Modifications**
  Obtaining timely feedback from the required official bodies (Traffic Police, RTO, Smart City officials, etc.) involves long bureaucratic formalities and waiting times, delaying necessary modifications and performance optimisations

References

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